

Math 133 Written Homework 1 Solutions

1. A product launch, viewed through functions.

The data are

w	1	2	3	4	5	6
$P(w)$	18	20	22	20	19	21
$Q(w)$	420	380	330	395	410	360

- (a) The domains of both functions are the six weeks

$$\text{dom}(P) = \text{dom}(Q) = \{1, 2, 3, 4, 5, 6\} \text{ weeks.}$$

Their ranges are

$$\text{range}(P) = \{18, 19, 20, 21, 22\} \text{ dollars per unit}$$

and

$$\text{range}(Q) = \{330, 360, 380, 395, 410, 420\} \text{ units.}$$

- (b) Each week is assigned exactly one price and exactly one quantity, so P and Q are functions of w . However, Q is not a function of P : the price was 20 in both weeks 2 and 4, but the corresponding quantities were 380 and 395.

- (c) Multiplying price by quantity gives

w	1	2	3	4	5	6
$R(w)$	7560	7600	7260	7900	7790	7560

in dollars. Thus

$$\text{dom}(R) = \{1, 2, 3, 4, 5, 6\} \text{ weeks}$$

and

$$\text{range}(R) = \{7260, 7560, 7600, 7790, 7900\} \text{ dollars.}$$

Revenue was largest in week 4, when it was \$7,900.

- (d) The claim is false. For example, the week-1 price of \$18 was lower than the week-2 price of \$20, but week-1 revenue, \$7,560, was also lower than week-2 revenue, \$7,600.
- (e) R is not a function of P for these data. The input price 20 would have to produce both $R(2) = 7600$ and $R(4) = 7900$. The table also cannot establish causation because other factors may have changed across the six weeks, such as advertising, customer demand, seasonality, or product availability.

2. The average cost of a market-day candle.

Here

$$C(q) = 180 + 3.60q, \quad A(q) = \frac{C(q)}{q} = 3.60 + \frac{180}{q}.$$

- (a) The input of both functions is q , the number of candles produced. The output $C(q)$ is total cost in dollars, and $A(q)$ is average cost in dollars per candle. Since candles are counted in whole numbers and at most 120 can be made,

$$\text{dom}_{\text{context}}(C) = \{0, 1, 2, \dots, 120\}.$$

- (b) We have $C(0) = 180$, the booth cost incurred when no candles are made. In contrast, $A(0)$ requires division by zero and is undefined. The natural algebraic domain is

$$\text{dom}_{\text{alg}}(A) = \mathbb{R} \setminus \{0\},$$

while the contextual domain is

$$\text{dom}_{\text{context}}(A) = \{1, 2, \dots, 120\}.$$

- (c)

$$A(12) = 3.60 + \frac{180}{12} = 18.60, \quad A(60) = 3.60 + \frac{180}{60} = 6.60.$$

Producing 12 candles gives an average cost of \$18.60 per candle, while producing 60 gives an average cost of \$6.60 per candle.

- (d) The exact contextual range is

$$\left\{ 3.60 + \frac{180}{q} : q \in \mathbb{Z}, 1 \leq q \leq 120 \right\} \quad \text{dollars per candle.}$$

Because $3.60 + 180/q$ decreases as positive q increases, its maximum is

$$A(1) = 183.60$$

and its minimum is

$$A(120) = 5.10.$$

- (e) For $q > 0$,

$$3.60 + \frac{180}{q} \leq 6 \iff \frac{180}{q} \leq 2.40 \iff q \geq 75.$$

Therefore, the smallest feasible production quantity is 75 candles.

3. A two-tier delivery model.

- (a) The contextual domain is $[0, 20]$ miles. On the first interval, prices run from 7.50 to 13.50 dollars. On the second interval, they run from just above 13.50 through 22.50 dollars. Together,

$$\text{range}(D) = [7.50, 22.50] \text{ dollars.}$$

(b)

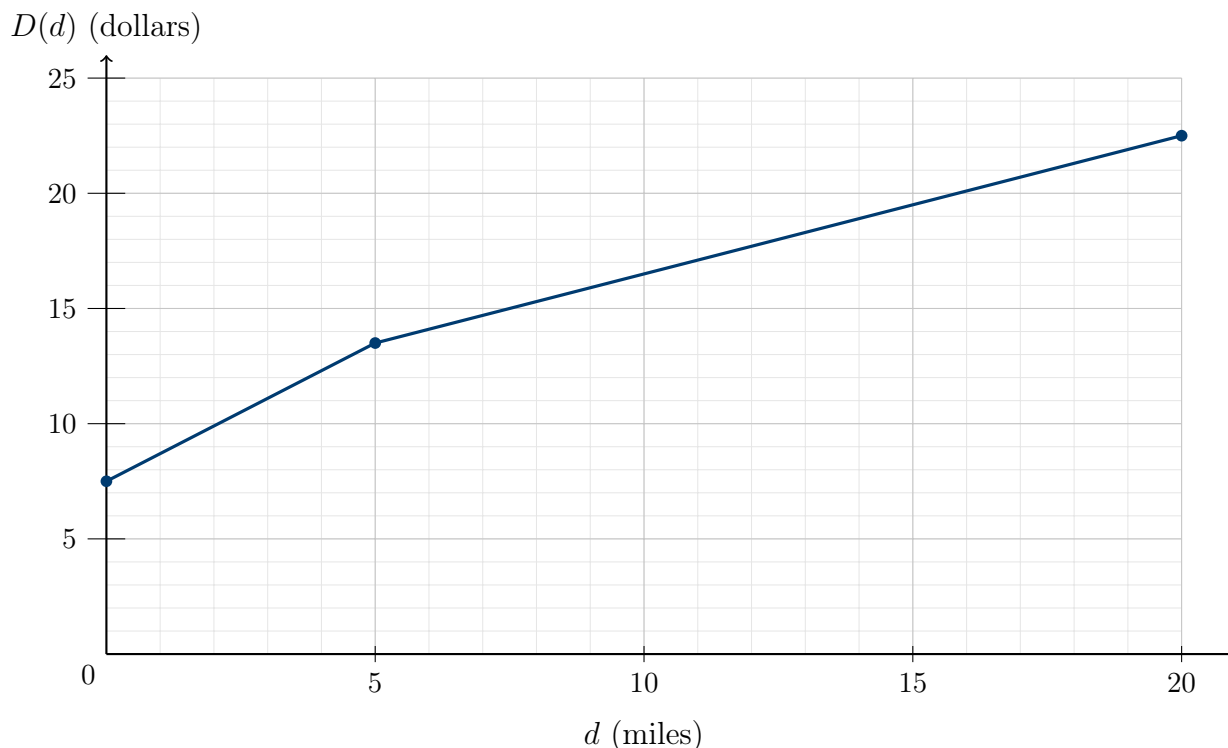
$$D(0) = 7.50, \quad D(5) = 7.50 + 1.20(5) = 13.50,$$

and, since $12 > 5$,

$$D(12) = 10.50 + 0.60(12) = 17.70.$$

Thus the pre-surcharge prices are \$7.50 for a zero-mile delivery, \$13.50 for a five-mile delivery, and \$17.70 for a twelve-mile delivery.

- (c) The graph consists of the line segment from $(0, 7.50)$ to $(5, 13.50)$ followed by the line segment from $(5, 13.50)$ to $(20, 22.50)$. All three displayed endpoints are included in the graph. Since both slopes are positive, D is increasing on its domain.



- (d) At the five-mile boundary, the first formula gives

$$7.50 + 1.20(5) = 13.50,$$

and the second formula would give

$$10.50 + 0.60(5) = 13.50.$$

The values agree, so there is no price jump. What changes is the rate per additional mile: the slope decreases from \$1.20 per mile to \$0.60 per mile.

- (e) Multiplying each branch by 1.0775 gives

$$F(d) = \begin{cases} 8.08125 + 1.293d, & 0 \leq d \leq 5, \\ 11.31375 + 0.6465d, & 5 < d \leq 20. \end{cases}$$

For a twelve-mile delivery,

$$F(12) = 1.0775D(12) = 1.0775(17.70) = 19.07175.$$

The final charge is therefore \$19.07.

4. Spreadsheet investigation: 30 daily S&P 500 values.

- (a) The downloaded file has headings `observation_date` and `SP500` in row 1 and 30 observations in rows 2–31. Column A should be formatted as dates and column B as numbers with two decimal places. The first observation is July 21, 2026, with index level 7509.20; the last is August 31, 2026, with index level 7686.14.

- (b) The required entries are

$$C2 = N/A, \quad D2 = N/A,$$

$$C3 = B3-B2, \quad D3 = (B3-B2)/B2, \quad E2 = 100*B2/ \$B\$2.$$

Quick Fill copies the first two formulas through row 31 and the third through row 31. Cell E2 is 100.00. An index value of 102 means that the S&P 500 value on that date is 2% greater than its July 21 value.

The completed calculations, rounded only for display, are shown below.

Date	S&P 500	Point change	Percent change	Value index
2026-07-21	7509.20	—	—	100.00
2026-07-22	7498.96	-10.24	-0.14%	99.86
2026-07-23	7408.30	-90.66	-1.21%	98.66
2026-07-24	7411.98	3.68	0.05%	98.71
2026-07-27	7413.18	1.20	0.02%	98.72
2026-07-28	7428.78	15.60	0.21%	98.93
2026-07-29	7316.15	-112.63	-1.52%	97.43
2026-07-30	7437.63	121.48	1.66%	99.05
2026-07-31	7489.72	52.09	0.70%	99.74
2026-08-03	7600.50	110.78	1.48%	101.22
2026-08-04	7736.52	136.02	1.79%	103.03
2026-08-05	7723.55	-12.97	-0.17%	102.85
2026-08-06	7709.96	-13.59	-0.18%	102.67
2026-08-07	7757.64	47.68	0.62%	103.31
2026-08-10	7753.11	-4.53	-0.06%	103.25
2026-08-11	7728.20	-24.91	-0.32%	102.92
2026-08-12	7748.50	20.30	0.26%	103.19
2026-08-13	7798.99	50.49	0.65%	103.86
2026-08-14	7785.76	-13.23	-0.17%	103.68
2026-08-17	7745.06	-40.70	-0.52%	103.14
2026-08-18	7691.76	-53.30	-0.69%	102.43
2026-08-19	7707.98	16.22	0.21%	102.65
2026-08-20	7641.16	-66.82	-0.87%	101.76
2026-08-21	7674.37	33.21	0.43%	102.20
2026-08-24	7652.86	-21.51	-0.28%	101.91
2026-08-25	7677.28	24.42	0.32%	102.24
2026-08-26	7675.70	-1.58	-0.02%	102.22
2026-08-27	7730.99	55.29	0.72%	102.95
2026-08-28	7711.76	-19.23	-0.25%	102.70
2026-08-31	7686.14	-25.62	-0.33%	102.36

(c) The spreadsheet formulas give

$$\text{AVERAGE}(B2 : B31) = 7628.389666 \dots \approx 7628.39,$$

$$\text{MIN}(B2 : B31) = 7316.15 \quad \text{on July 29, 2026,}$$

and

$$\text{MAX}(B2 : B31) = 7798.99 \quad \text{on August 13, 2026.}$$

The total point change is

$$\text{SUM}(C3 : C31) = 176.94 \text{ points.}$$

As a check,

$$B31 - B2 = 7686.14 - 7509.20 = 176.94 \text{ points.}$$

- (d) The two completed graphs appear below. One valid observation from the first graph is that the index fell through July 29, then rose and reached its highest displayed value on August 13. One valid observation from the second graph is that the largest positive daily percentage change occurred on August 4, while the largest negative daily percentage change occurred on July 29.

